

STAT 24410/30030, Homework 3
Due Thursday, October 24, at 2:00pm

1. (15 pts) Lecture Notes, Chapter 2, Exercise 12.

2. (15 pts) Lecture Notes, Chapter 3, Exercise 2.

3. (15 pts) Lecture Notes, Chapter 3, Exercise 4.

4. (20 pts) Lecture Notes, Chapter 3, Exercise 6.

5. (15 pts) In Lecture 6 on Thursday 10/17, we talked about two estimators for the parameter of the exponential distribution. The setting is as follows: suppose that $T_1, \dots, T_n$ are independent and that each is distributed $\text{Exp}(\lambda)$. We denote with $\bar{T}_n$ the sample mean based on these observations and we introduced:

$$\hat{\lambda}_1 = \frac{1}{\bar{T}_n} = \frac{n}{\sum T_i},$$

and

$$\hat{\lambda}_2 = \sqrt{\frac{2n}{\sum T_i^2}}.$$

(i) Calculate the variance of T^2 where T is distributed $\text{Exp}(\lambda)$.

(ii) What is the central limit theorem for $\sum T_i^2/n$?

(iii) Using the delta method, derive the corresponding convergence in distribution result for $\hat{\lambda}_2$.

Recall that we derived this for $\hat{\lambda}_1$ in class.

6. (20 pts) In this exercise, you will explore empirically the two estimators from Exercise 5: $\hat{\lambda}_1$ and $\hat{\lambda}_2$. Provide all code you use.

(i) Choose $\lambda = 3$ and generate 10000 samples of 10 random variables $T_1, \dots, T_{10}$ with distribution $\text{Exp}(3)$. For each of the 10000 samples, construct the two estimators for λ . This should give you 10000 pairs of estimators for λ , each of them constructed using 10 independent draws. Based on these simulations estimate the bias, the variance, and the mean squared errors for each of the two estimators. How do they compare? Which estimator would you choose based on this analysis?

(ii) Plot two histograms: first, the 10000 $\hat{\lambda}_1$ estimators you obtained in (i); second, the 10000 $\hat{\lambda}_2$ estimators you obtained in (i). Do the histograms agree with the asymptotic distribution of the estimators we derived in class and in Exercise 5?